4.5 Notes and References

The literature on robot kinematics is quite extensive, and with very few exceptions most approaches are based on the Denavit–Hartenberg (D–H) parameters originally presented in [34] and summarized in Appendix C. Our approach is based on the Product of Exponentials (PoE) formula first presented by Brockett in [20]. Computational aspects of the PoE formula are discussed in [132].

Appendix C also elucidates in some detail the many advantages of the PoE formula over the D–H parameters, e.g., the elimination of link reference frames, the uniform treatment of revolute and prismatic joints, and the intuitive geometric interpretation of the joint axes as screws. These advantages more than offset the lone advantage of the D–H parameters, namely that they constitute a minimal set. Moreover, it should be noted that when using D–H parameters, there are differing conventions for assigning link frames, e.g., some methods align the joint axis with the $\hat{\mathrm{x}}$-axis rather than the $\hat{\mathrm{z}}$-axis of the link frame as we have done. Both the link frames and the accompanying D–H parameters need to be specified together in order to have a complete description of the robot's forward kinematics.

In summary, unless using a minimal set of parameters to represent a joint's spatial motion is critical, there is no compelling reason to prefer the D–H parameters over the PoE formula. In the next chapter, an even stronger case can be made for preferring the PoE formula to model the forward kinematics.

4.6 Exercises

Exercise 4.1 Familiarize yourself with the functions `FKinBody` and `FKinSpace` in your favorite programming language. Can you make these functions more computationally efficient? If so, indicate how. If not, explain why not.

Exercise 4.2 The RRRP SCARA robot of Figure 4.12 is shown in its zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{\mathrm{b}\}$. For $\ell_0 = \ell_1 = \ell_2 = 1$ and the joint variable values $\theta = (0, \pi/2, -\pi/2, 1)$, use both the `FKinSpace` and the `FKinBody` functions to find the end-effector configuration $T \in SE(3)$. Confirm that they agree with each other.

Exercise 4.3 Determine the end-effector frame screw axes $\mathcal{B}_i$ for the 3R robot

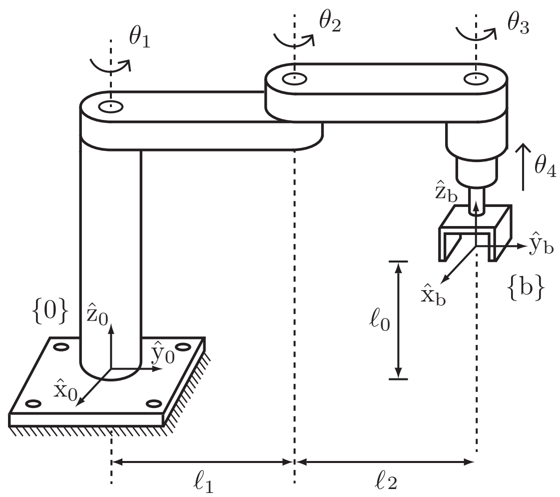

Figure 4.12: An RRRP SCARA robot for performing pick-and-place operations.

in Figure 4.3.

Exercise 4.4 Determine the end-effector frame screw axes $\mathcal{B}_i$ for the RRPRRR robot in Figure 4.5.

Exercise 4.5 Determine the end-effector frame screw axes $\mathcal{B}_i$ for the UR5 robot in Figure 4.6.

Exercise 4.6 Determine the space frame screw axes $\mathcal{S}_i$ for the WAM robot in Figure 4.8.

Exercise 4.7 The PRRRRR spatial open chain of Figure 4.13 is shown in its zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$.

Exercise 4.8 The spatial RRRRPR open chain of Figure 4.14 is shown in its zero position, with fixed and end-effector frames chosen as indicated. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$.

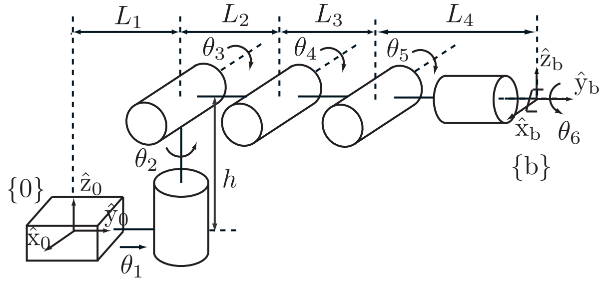

Figure 4.13: A PRRRRR spatial open chain at its zero configuration.

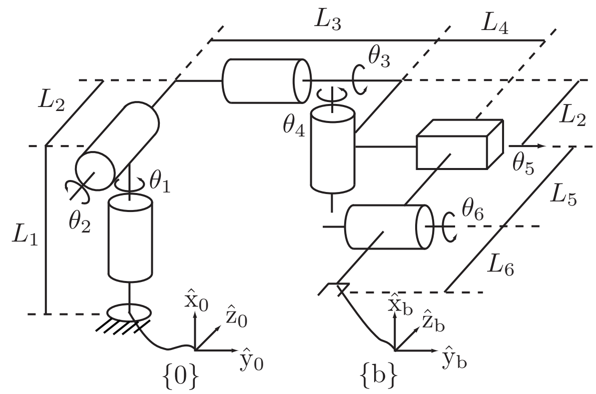

Figure 4.14: A spatial RRRRPR open chain.

Exercise 4.9 The spatial RRPPRR open chain of Figure 4.15 is shown in its zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$.

Exercise 4.10 The URRPR spatial open chain of Figure 4.16 is shown in its zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$.

Exercise 4.11 The spatial RPRRR open chain of Figure 4.17 is shown in its

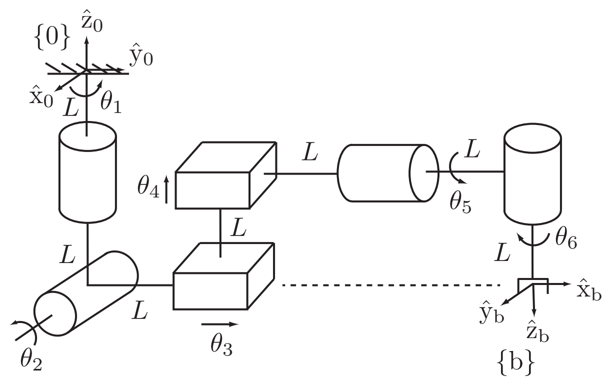

Figure 4.15: A spatial RRPPRR open chain with prescribed fixed and end-effector frames.

zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$.

Exercise 4.12 The RRPRRR spatial open chain of Figure 4.18 is shown in its zero position (all joints lie on the same plane). Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$. Setting $\theta_5 = \pi$ and all other joint variables to zero, find T_{06} and T_{60}.

Exercise 4.13 The spatial RRRPRR open chain of Figure 4.19 is shown in its zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$.

Exercise 4.14 The RPH robot of Figure 4.20 is shown in its zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{s\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$. Use both the `FKinSpace` and the `FKinBody` functions to find the end-effector configuration $T \in SE(3)$ when $\theta = (\pi/2, 3, \pi)$. Confirm that the results agree.

Exercise 4.15 The HRR robot in Figure 4.21 is shown in its zero position. Determine the end-effector zero position configuration M, the screw axes $\mathcal{S}_i$ in $\{0\}$, and the screw axes $\mathcal{B}_i$ in $\{b\}$.

Exercise 4.16 The forward kinematics of a four-dof open chain in its zero

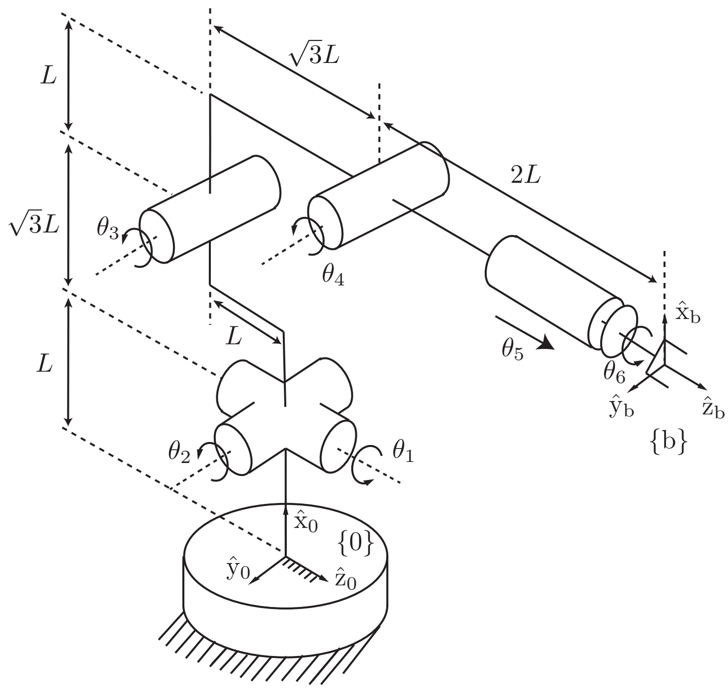

Figure 4.16: A URRPR spatial open-chain robot.

position is written in the following exponential form:

$$T(\theta) = e^{[A_1]\theta_1} e^{[A_2]\theta_2} M e^{[A_3]\theta_3} e^{[A_4]\theta_4}.$$

Suppose that the manipulator's zero position is redefined as follows:

$$(\theta_1, \theta_2, \theta_3, \theta_4) = (\alpha_1, \alpha_2, \alpha_3, \alpha_4).$$

Defining ${\theta'}_i = \theta_i - \alpha_i, i = 1, \ldots, 4$, the forward kinematics can then be written

$$T_{04}({\theta'}_1, {\theta'}_2, {\theta'}_3, {\theta'}_4) = e^{[{A'}_1]{\theta'}_1} e^{[{A'}_2]{\theta'}_2} M' e^{[{A'}_3]{\theta'}_3} e^{[{A'}_4]{\theta'}_4}.$$

Find M' and each of the ${A'}_i$.

Exercise 4.17 Figure 4.22 shows a snake robot with end-effectors at each end. Reference frames $\{b_1\}$ and $\{b_2\}$ are attached to the two end-effectors, as shown.

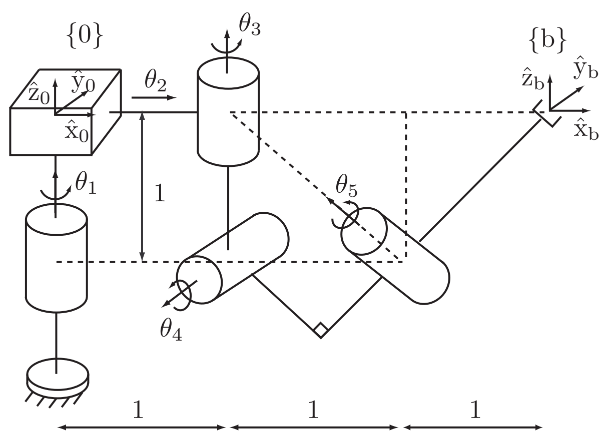

Figure 4.17: An RPRRR spatial open chain.

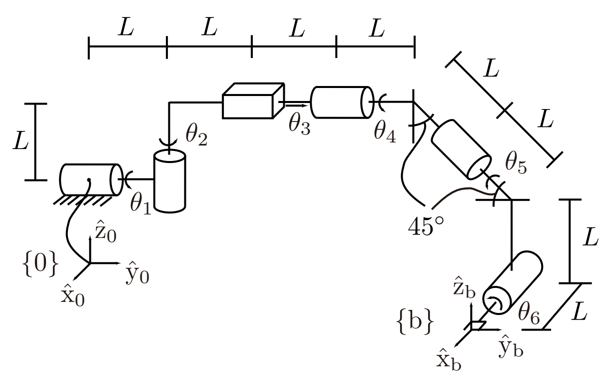

Figure 4.18: An RRPRRR spatial open chain.

(a) Suppose that end-effector 1 is grasping a tree (which can be thought of as "ground") and end-effector 2 is free to move. Assume that the robot is in its zero configuration. Then $T_{b_1 b_2} \in SE(3)$ can be expressed in the following product of exponentials form:

$$T_{b_1 b_2} = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} \cdots e^{[\mathcal{S}_5]\theta_5} M.$$

Find $\mathcal{S}_3, \mathcal{S}_5$, and M.

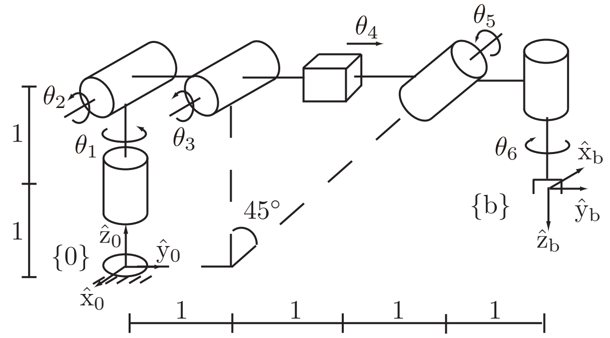

Figure 4.19: A spatial RRRPRR open chain with prescribed fixed and end-effector frames.

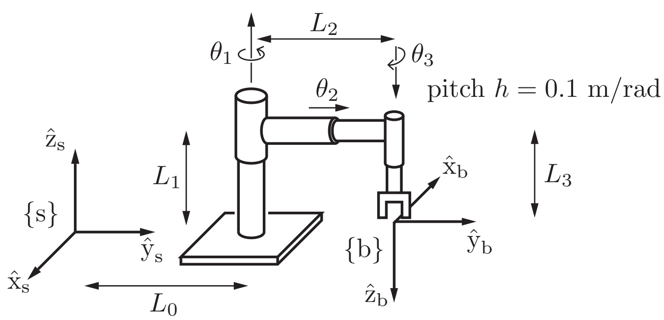

Figure 4.20: An RPH open chain shown at its zero position. All arrows along/about the joint axes are drawn in the positive direction (i.e., in the direction of increasing joint value). The pitch of the screw joint is 0.1 m/rad, i.e., it advances linearly by 0.1 m for every radian rotated. The link lengths are $L_0 = 4$, $L_1 = 3$, $L_2 = 2$, and $L_3 = 1$ (figure not drawn to scale).

(b) Now suppose that end-effector 2 is rigidly grasping a tree and end-effector 1 is free to move. Then $T_{b_2 b_1} \in SE(3)$ can be expressed in the following product of exponentials form:

$$T_{b_2 b_1} = e^{[\mathcal{A}_5]\theta_5} e^{[\mathcal{A}_4]\theta_4} e^{[\mathcal{A}_3]\theta_3} N e^{[\mathcal{A}_2]\theta_2} e^{[\mathcal{A}_1]\theta_1}.$$

Find $\mathcal{A}_2, \mathcal{A}_4$, and N.

Exercise 4.18 The two identical PUPR open chains of Figure 4.23 are shown

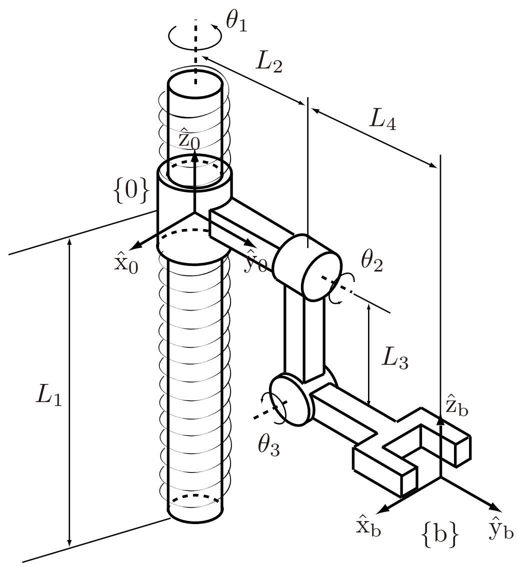

Figure 4.21: HRR robot. The pitch of the screw joint is denoted by h.

in their zero position.

(a) In terms of the given fixed frame {A} and end-effector frame {a}, the forward kinematics for the robot on the left (robot A) can be expressed in the following product of exponentials form:

$$T_{Aa} = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} \cdots e^{[\mathcal{S}_5]\theta_5} M_a.$$

Find $\mathcal{S}_2$ and $\mathcal{S}_4$.

(b) Suppose that the end-effector of robot A is inserted into the end-effector of robot B in such a way that the origins of the end-effectors coincide; the two robots then form a single-loop closed chain. Then the configuration space of the single-loop closed chain can be expressed in the form

$$M = e^{-[\mathcal{B}_5]\phi_5} e^{-[\mathcal{B}_4]\phi_4} e^{-[\mathcal{B}_3]\phi_3} e^{-[\mathcal{B}_2]\phi_2} e^{-[\mathcal{B}_1]\phi_1} e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} e^{[\mathcal{S}_4]\theta_4} e^{[\mathcal{S}_5]\theta_5}$$

for some constant $M \in SE(3)$ and $\mathcal{B}_i = (\omega_i, v_i)$, for $i = 1, \ldots, 5$. Find $\mathcal{B}_3$, $\mathcal{B}_5$, and M. (Hint: Given any $A \in \mathbb{R}^{n \times n}$, $(e^A)^{-1} = e^{-A}$).

Exercise 4.19 The RRPRR spatial open chain of Figure 4.24 is shown in its zero position.

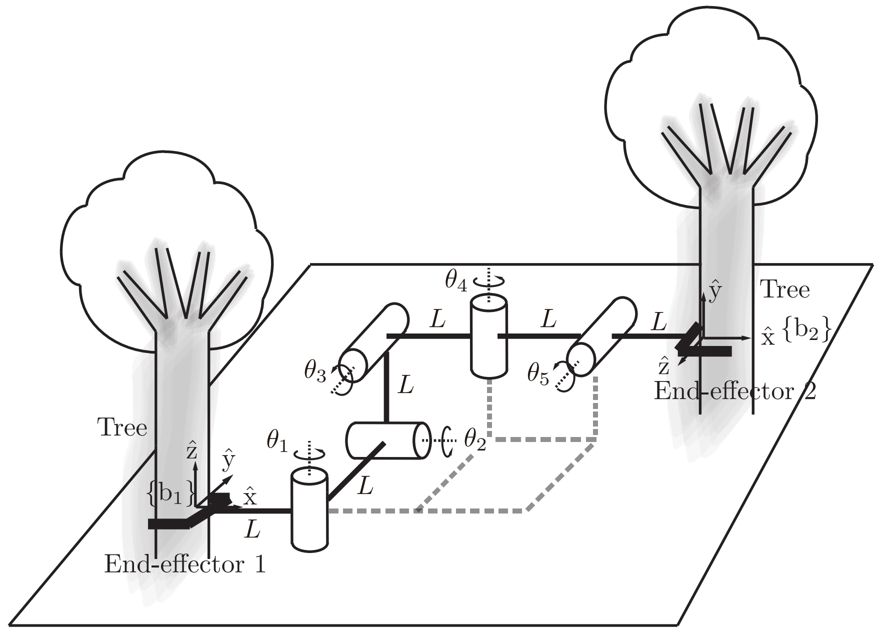

Figure 4.22: Snake robot.

(a) The forward kinematics can be expressed in the form

$$T_{sb} = M_1 e^{[\mathcal{A}_1]\theta_1} M_2 e^{[\mathcal{A}_2]\theta_2} \cdots M_5 e^{[\mathcal{A}_5]\theta_5}.$$

Find $M_2, M_3, \mathcal{A}_2$, and $\mathcal{A}_3$. (Hint: Appendix C may be helpful.)

(b) Expressing the forward kinematics in the form

$$T_{sb} = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} \cdots e^{[\mathcal{S}_5]\theta_5} M,$$

find M and $\mathcal{S}_1, \ldots, \mathcal{S}_5$ in terms of the quantities $M_1, \ldots, M_5$, $\mathcal{A}_1, \ldots, \mathcal{A}_5$ appearing in (a).

Exercise 4.20 The spatial PRRPRR open chain of Figure 4.25 is shown in its zero position, with space and end-effector frames chosen as indicated. Derive its forward kinematics in the form

$$T_{0n} = e^{[S_1]\theta_1} e^{[S_2]\theta_2} e^{[S_3]\theta_3} e^{[S_4]\theta_4} e^{[S_5]\theta_5} M e^{[S_6]\theta_6},$$

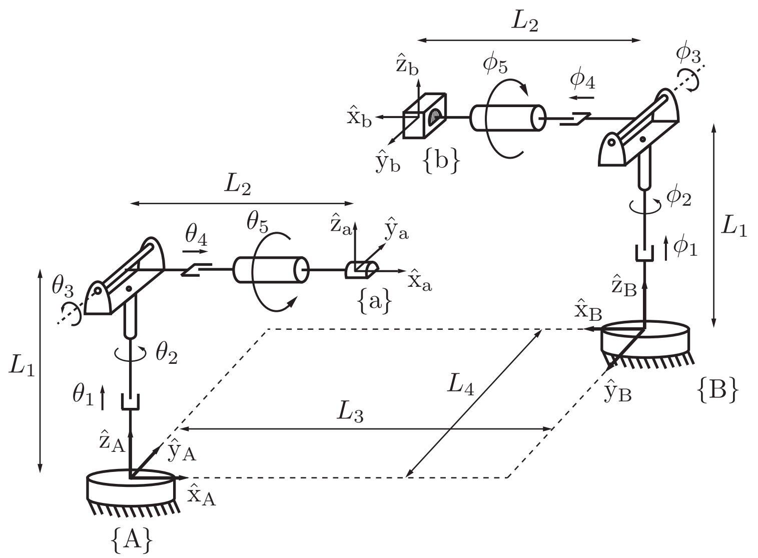

Figure 4.23: Two PUPR open chains.

where $M \in SE(3)$.

Exercise 4.21 (Refer to Appendix C.) For each $T \in SE(3)$ below, find, if they exist, values for the four parameters (α, a, d, ϕ) that satisfy

$$T = \mathrm{Rot}(\hat{\mathrm{x}}, \alpha)\,\mathrm{Trans}(\hat{\mathrm{x}}, a)\,\mathrm{Trans}(\hat{\mathrm{z}}, d)\,\mathrm{Rot}(\hat{\mathrm{z}}, \phi).$$

(a) $T = \begin{bmatrix} 0 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$

(b) $T = \begin{bmatrix} \cos\beta & \sin\beta & 0 & 1 \\ \sin\beta & -\cos\beta & 0 & 0 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$

(c) $T = \begin{bmatrix} 0 & -1 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$

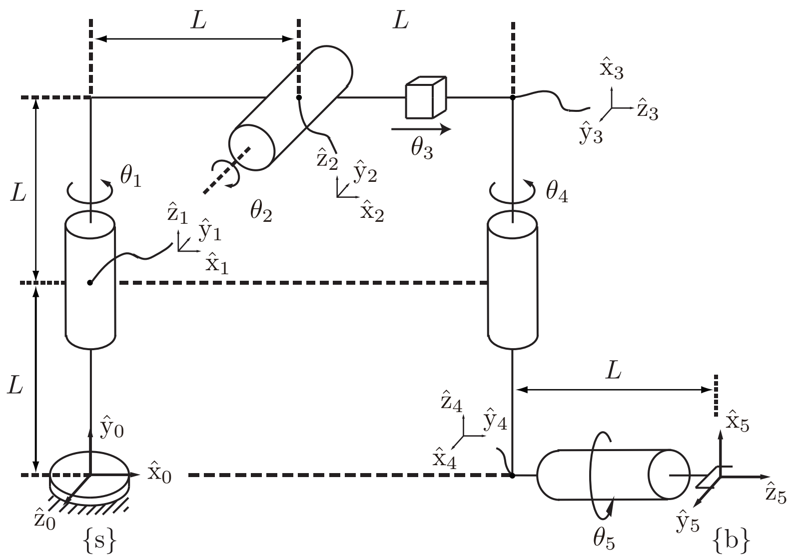

Figure 4.24: A spatial RRPRR open chain.

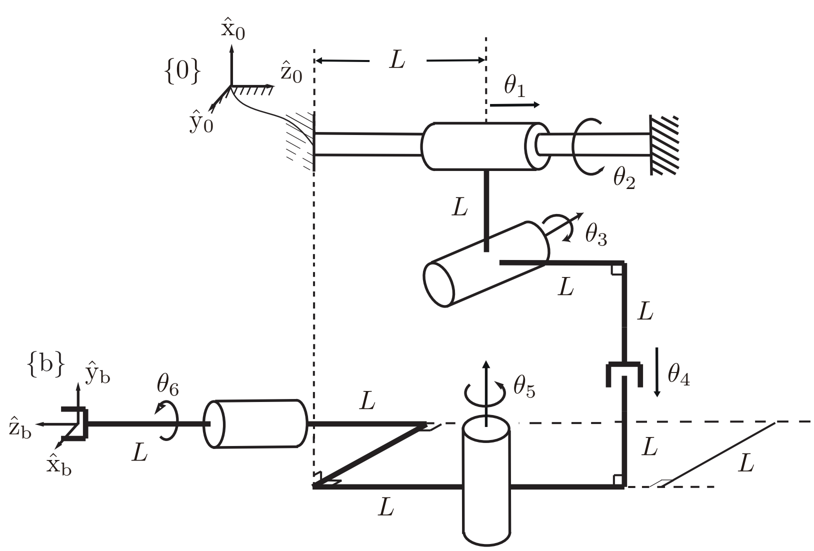

Figure 4.25: A spatial PRRPRR open chain.